The Caseworker Will See You Now

A Structural Analysis of Dynamic Sickness Leave Behavior *

Sophie Laura Bune†

April 6, 2026

Abstract

*Number of characters:

†University of Copenhagen, Department of Economics, study number QRN838.

Contents

1 Introduction **1**
1.1 Relevant literature . . . 1
1.1.1 Structural dynamic models with health shocks . . . 1
1.1.2 Reduced form results on sickness leave dynamics . . . 2
1.1.3 Dynamic UI models . . . 2

2 Data and institutional setting **2**
2.1 Danish sickness leave and unemployment insurance policy . . . 2
2.2 Data . . . 3
2.3 Descriptive analysis . . . 4
2.4 Reduced form results . . . 4

3 Model **4**
3.1 State variables and transition rules . . . 4
3.2 Income and benefits . . . 5
3.2.1 Employment . . . 5
3.2.2 Unemployment . . . 6
3.2.3 Sickness leave . . . 6
3.3 Value functions . . . 7
3.4 Solve . . . 8
3.5 Simulate . . . 8

4 Structural estimation **9**
4.1 Identification . . . 9
4.2 Results . . . 9

5 Discussion and concluding remarks **10**

6 Appendix **12**

List of Figures

1 UI benefits across durations when starting a spell with full eligibility . . . 3
2 Model flow to and from employment, unemployment and sickness leave . . . 8
3 First model results . . . 9
4 First results . . . 10

List of Tables

1 Introduction

What are the behavioral effects of the reassessment timing?
What are the behavioral effects of changes in the UI system on sickness leave behvior and the other way around?

Counterfactual policy analysis

Given estimated parameters, simulate what would happen under alternative policies:
Earlier or later reassessment (t_reassess)
Stricter medical gate (delta0_doc, delta1_doc)
Lower replacement rate during sick leave
Shorter maximum UI duration (Ubar)
Synergy effects on the UI (sickness leave) system from changing the sickness leave (UI) system
Estimating the effect on average duration of reassessment timing in each month
Effects on effectivity and wages when cutting in the sickness leave

1.1 Relevant literature

1.1.1 Structural dynamic models with health shocks

Capatina and Keane (2023): Life-cycle model with endogenous wages, human capital, and employer-sponsored health insurance. Shows that health shocks drive earnings inequality, and that providing public insurance to the uninsured improves health and labor supply enough to largely offset program costs.

Jeske and Kitao (2009): Dynamic GE model asking whether the regressive US health insurance tax subsidy (only for employer-sponsored plans) can be improved. Finds that removing it risks collapsing the group insurance market, but extending a refundable credit to individual plans could improve both coverage and welfare.

Khwaja (2010): Estimates willingness to pay for Medicare using a structural life-cycle model with endogenous health production. Finds average WTP around $25k (1991 $), with less-educated individuals valuing it more. Medicare's main benefit is insurance against expenditure risk, not longevity gains.

De Nardi et al. (2025): Shows that health dynamics are driven largely by fixed health types (predetermined early in life), not just shocks. Incorporating this into a structural model, they find that lifetime monetary and welfare costs of bad health are large and concentrated — mostly income losses for working-age people.

Kitao (2014): GE life-cycle model with search frictions, health uncertainty, and disability insurance (DI). Finds that Medicare benefits within DI are crucial — removing them would sharply

reduce DI take-up. Also shows DI enrollment amplifies the effects of Social Security and Medicare reforms.

French (2005): Structural life cycle model of retirement that separates health effects through wages, medical costs, time endowment, and survival. Finds all four channels matter for retirement timing, with health affecting labor supply both directly and through wealth accumulation. Models health as binary: good or bad.

Capatina (2015): Calibrated life-cycle model decomposing health's effects through four channels: productivity, medical expenditure, time endowment, and survival. Productivity and time endowment dominate. Health risk explains much of the gap in labor supply and government transfer reliance between college and non-college workers.

Hosseini et al. (2022): The paper constructs a unified, objective measure of health called the frailty index — defined as the cumulative share of adverse health deficits an individual has accumulated (ADLs, medical diagnoses, mental health indicators). Using PSID data (2003–2017), they document four stylized facts about health dynamics: 1. Mean frailty increases with age, is higher for less-educated individuals, and slightly higher for women, 2. Cross-sectional dispersion in frailty rises with age, 3. The distribution of frailty is right-skewed — many people have zero deficits, with a long tail of high-frailty individuals, 4. Frailty is highly persistent — people are much more likely to stay in the same frailty quintile than move. They then estimate a stochastic AR(1) process for frailty with individual fixed effects and a transitory shock, correcting for selection bias due to mortality. The estimated process is designed to plug directly into quantitative life-cycle models.

1.1.2 Reduced form results on sickness leave dynamics

1.1.3 Dynamic UI models

2 Data and institutional setting

2.1 Danish sickness leave and unemployment insurance policy

The model is set up to approximately mimic the Danish sickness leave and UI systems. The Danish labor market is intended to be as flexible as possible, which requires strong public insurance systems to catch whomever is temporarily unable to work. The Danish universal welfare model thus entails voluntary insurance based UI benefits and universal sickness leave benefits.

To be eligible for UI benefits the unemployed is required to have been a member of an unemployment fund for at least one year prior to the unemployment spell. Furthermore, they are required to have worked hours or earned a wage corresponding to at least one year of full time work over the course of the 3 years prior to the unemployment spell. The unemployed is required to register as searching for a job on the first day of unemployment and be available for the labor market their entirety of the spell, i.e. be able to take a job with a one day notice. A UI eligible worker is

entitled to two years of UI benefits over the course of three years. UI since 2017 have been used on an hourly basis. If the unemployed exits UI into employment but does not re-earn a new full UI eligibility by working one full time year, they will re-enter their previous UI spell by using what might be left of that eligibility. In principle, UI benefits are 90 pct. of the average of the wage, minus 8 percent in labor market contribution, during the 12 best months out of the past 24 months before the unemployment spell. But the maximum UI benefit is 22,041 DKK (∼2,950€) a month, which means that the majority of workers does not have a UI replacement rate of 90 percent. If the unemployed worker has earned "double eligibility", meaning two years as an unemployment fund member and two years of full time work in the past three years, and would have received the maximum UI benefit, then the worker will be eligible for the employment bonus. The employment bonus is an increased benefit the first three months of the unemployment spell such that the benefit will be increased to 26,620 DKK (∼3,550€) a month, conditional on that it is still lower than a 90 percent replacement rate. If the unemployed worker is still unemployment by the end of their UI eligibility, they will transition to social assistance if they are still an active job seeker and have assets for less than 16,200 DKK (∼2,150€). Social assistance corresponds to 13,060 DKK (∼1,750€) a month (see Figure 1).

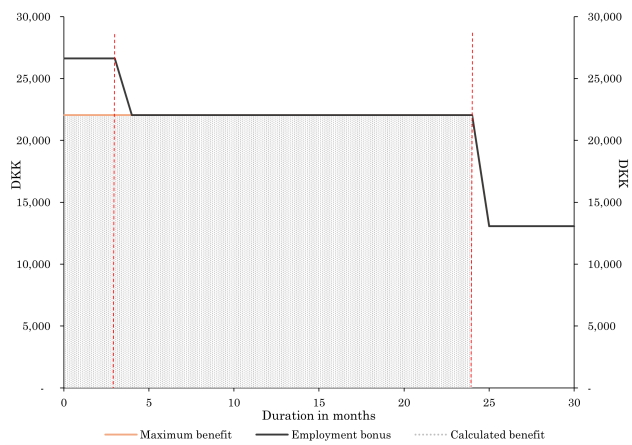

Figure 1: UI benefits across durations when starting a spell with full eligibility

Workers are eligible for sickness leave benefits if they are employed or unemployed and eligible for UI benefits. Dependent on their collective agreement workers will typically receive their wage during sickness and their employer will receive the sickness leave benefits as compensation. The first 30 days of sickness will be payed fully by the employer such that they are only compensated for long periods of sickness. Similarly UI benefit recipients receive UI benefits for the first 14 days of sickness payed by the unemployment fund and then start receiving sickness leave benefits. The sickness leave benefit corresponds directly to the UI benefit but without the employment bonus.

2.2 Data

I use data from Statistics Denmark. On Ui benefit recipients I use data on recipients of public transfers from where I join with data on wage earners to have both UI and employment data. I

then have the foundation to calculate UI benefit usage, as I will have to keep track on whether or not the unemployed has re-earned a new UI eligibility as explained above. The goal is to be able to estimate empirical hazard rates to match to the model hazard rates. I calculate the UI usage from 2010 and on wards to be able to calculate hazard rates in two reform periods 2010 to 2023 and 2023 to 2025.

2.3 Descriptive analysis

2.4 Reduced form results

3 Model

I model a work life with health shocks in a finite horizon, discrete time model where one time period corresponds to one month in a universal welfare state with public sickness leave benefits, UI and social assistance benefits, in this case the Danish policy system. The model has no absorbing states which allow the agents to freely transition between employment, unemployment and sickness leave. The employed workers have the binary choices to quit their job and enter unemployment and to enter sickness leave. The unemployed workers choose a continuous search effort that maps into their probability of entering employment the following month and also has the binary choice of transitioning to sickness leave. The agents on sickness leave only has the binary choice of exiting to their origin.

3.1 State variables and transition rules

The labor market status is treated in three different value functions and is thus not a true state, but notation wise I note them

$$j_t = \begin{cases} 2 & \text{Sickness leave} \\ 1 & \text{Employed} \\ 0 & \text{Unemployed.} \end{cases} \tag{1}$$

Origin of sickness leave determines how the agent can transition as someone on sick leave from unemployment cannot transition from sickness leave to employment. When they start a new sickness leave spell their origin becomes their last non-sickness leave employment state,

$$o_t = \begin{cases} j_{t-1} & \text{if } j_t = 2 \text{ and } d_t^S = 1 \\ o_{t-1} & \text{otherwise.} \end{cases} \tag{2}$$

The sick workers still have the risk of being fired during their sickness leave, in which case their origin changes from 1 to 0.

Human capital is accumulated as effective work hours based on the agent's health level to

represent the lower efficiency when sick,

$$k_{t+1} = (1-\omega)k_t + \mathbf{1}_{\{j_t=1\}}h_t. \tag{3}$$

Unemployment eligibility will be build up during employment and run down during unemployment. If the agent is unemployed, $j_t = 0$, and not UI eligible, $d_t^U = 0$, they will receive social assistance.

$$d_{t+1}^U = \begin{cases} d_t^U & \text{if } j_t = 2 \\ d_t^U + \zeta & \text{if } j_t = 1 \\ d_t^U - 1 & \text{if } j_t = 0, \end{cases} \tag{4}$$

where ζ depends on the current policy on re-qualification. This is not an exact formulation of the Danish UI system but rather an approximation to keep the state space as simple as possible, as it does not ensure that the workers has obtained a full new right when reentering unemployment.

On the other hand, sickness leave duration will be build up during sickness leave and reset after a period of employment. Again, is this an approximation of the true policy rule.

$$d_{t+1}^S = \begin{cases} d_t^S + 1 & \text{if } j_t = 2 \\ 0 & \text{else,} \end{cases} \tag{5}$$

I treat the first benefit, b_t^0, on both sickness leave and unemployment as a state to keep track of the benefit path. Likewise, on sickness leave the benefit tier after reassessment, r_t, is treated as a state to be able to solve with backwards induction. As the benefit level need to be known in the last period and as it is determined in reassessment i choose to track it as a state. Both b_t^0 and r_t are spell and individual specific states.

The health process is modeled as a latent variable in an AR(1) process, to make it a stochastic variable that considers last period's health, while also being bounded between 0 and 1. There is included a positive drift in the AR(1) process for when the agents are on sickness leave, to represent their health improving faster compared to working and job searching.

$$z_{t+1} = \rho z_t + \upsilon \mathbf{1}_{(j_{t+1}=2)} + \varepsilon_{t+1}, \quad \varepsilon \sim N(0, \sigma^2) \tag{6}$$

$$h_{t+1} = \frac{\exp(z_{t+1})}{1+\exp(z_{t+1})} \tag{7}$$

3.2 Income and benefits

3.2.1 Employment

Employed workers receive a wage that is dependent on their human capital (Keane, 2016),

$$w_t = w(1+\alpha k_t). \tag{8}$$

3.2.2 Unemployment

UI benefit is pinned down at spell start and is therefore also a state as earlier described. Among UI recipients, the high benefit with the employment bonus (beskæftigelsestillæg), b^{emp}, applies for the first J periods of the spell, which equivalently means the last J periods have not yet been reached and the UI eligibility is a full right.

$$b_t^U(w_t(k_t), d_t^U) = \begin{cases} b^{wel} & \text{if} \quad d_t^U = 0 \\ b^{emp} & \text{if} \quad d_t^U > \bar{U} - J \\ \min(0.8\, w_t(k_t), b^{max}) & \text{if} \quad 0 < d_t^U \leq \bar{U} - J. \end{cases} \tag{9}$$

Thus, UI spells are characterized by a regular benefit level $b^{UI}(w_t(k_t))$ throughout the eligibility window, except that a subset of UI-eligible workers receive a temporarily higher benefit for J initial periods. This could have been modeled closer to the rules by adding a probability of receiving the short term higher benefit conditional on human capital equivalent to the probability of being eligible.

3.2.3 Sickness leave

Sickness leave benefits The sickness benefit, b_t, is at one level for all until the reassessment date, t_r. At this time there is a risk of reduction in benefit, to a lower level or 0 in benefit. The risk δ, depends on the state of health as terminally ill and other serious illnesses are more likely to not be reduced in benefit. The probability of being reduced in benefit is assumed to be a quadratic function of the health state,

$$\delta_t^{low}(h_t) = \delta_0^{lower} + \delta_1^{lower} h_t + \delta_2^{lower} h_t^2 \tag{10}$$

$$\delta_t^{out}(h_t) = \delta_0^{out} + \delta_1^{out} h_t + \delta_2^{out} h_t^2 \tag{11}$$

$$\delta_t^{high}(h_t) = 1 - \delta_t^{lower} - \delta_t^{out}. \tag{12}$$

Where δ_0 is the baseline risk and δ_1 is the additional likelihood of being reduced in benefit as health improves, and δ_2 is the curvature of the risk. The quadratic form is to be able to have that; at high levels of health it is more likely to be reduced to 0 in benefit, while at medium health levels it is more likely to be reduced to the lower benefit. The benefit at time t is

$$b_t^S = \begin{cases} b^{high} = b^{UI}, & \text{if} \quad d_t^S < t_r \\ \tilde{b}, & \text{if} \quad d_t^S \geq t_r \end{cases} \tag{13}$$

where $\tilde{b}$ is a stochastic variable that is b^{high} with probability $\delta_t^{high}(h_t)$, b^{low} with probability δ_t^{low}, and 0 with probability δ_t^{out}. Benefit path on sickness leave is treated as a state to solve the model via backwards induction. I also have solved a version of the model where the sick workers receive their wage during sickness leave but do not accumulate human capital.

Initial assessment is included to not make sickness leave much more attractive than work or job search as the decision maker does not have the cost of search or work when on sickness leave. It is modeled as the reassessment with a probability of not entering sickness leave that is increasing in health,

$$p_{\text{ini}}^{\text{out}}(h_t) = \delta_0^{\text{ini}} + \delta_1^{\text{ini}} h_t + \delta_2^{\text{ini}} h_t^2. \tag{14}$$

When an agent chooses entry into sickness leave, $g_t = 1$, value of sickness leave is

$$\tilde{V}_{t+1}^S = p_{\text{ini}}^{\text{out}}(h_t)\left[u(0) + \beta V_{t+2}^{o_t}\right] + \left(1 - p_{\text{ini}}^{\text{out}}(h_t)\right) V_{t+1}^S \tag{15}$$

3.3 Value functions

Value of being unemployed I changed the utility of consumption to match the form of disutility of search effort and Keane (2016).

$$\begin{aligned} V_t^U(h_t, k_t, d_t^U, b_t^0) \quad = \quad & \max_{s_t \in [\bar{s}\cdot \mathbf{1}_{(d_t>0)}, 1], g_t \in \{0,1\}} \Bigg\{ \eta \frac{c_t^{1+\mu}}{1+\mu} - (1-h_t)\left(\chi + \lambda_n \frac{s_t^{1+\gamma}}{1+\gamma}\right) \\ & + \beta E\big[(1-g_t)\big(s_t V_{t+1}^E(h_{t+1}, k_{t+1}, d_{t+1}^U) \\ & + (1-s_t) V_{t+1}^U(h_{t+1}, k_{t+1}, d_{t+1}^U, b_t^0)\big) \\ & + g_t \tilde{V}_{t+1}^S(h_{t+1}, k_{t+1}, d_{t+1}^U, d_{t+1}^S, r_{t+1}, 0, b_{t+1}^0) \big| h_t\big] \Bigg\} \end{aligned} \tag{16}$$

$$\text{s.t.}$$

$$c_t \quad = \quad y_t = (1-\tau) b_t^U(w_t(k_t), d_t^U), \tag{17}$$

where the search effort s_t directly translates to the probability of being employed the following period. The search cost is modeled like in DellaVigna et al. (2017), and $n \in \{1, 2\}$ represents the types of unobserved heterogeneity in the level of search cost. The UI-eligible unemployed face a minimum search requirement of $\bar{s}$ to make unemployment less attractive. In reality this search requirement cannot be perfectly monitored and should in fact be modeled as a probability of getting caught.

Value of being employed When employed, individuals earn wage income, face exogenous job separation risk σ and choose endogenous job separation, q, and entry into sickness leave, g

$$\begin{aligned} V_t^E(h_t, k_t, d_t^U) \quad = \quad & \max_{q_t, g_t \in \{0,1\}} \Bigg\{ \eta \frac{c_t^{1+\mu}}{1+\mu} - \nu_n \frac{(1-h_t)^{1+\iota}}{1+\iota} + \beta E\Bigg[(1-g_t)\Big(q_t V_{t+1}^U(h_{t+1}, k_{t+1}, d_{t+1}^U, b_t^0) \\ & + (1-q_t)\left((1-\sigma) V_{t+1}^E(h_{t+1}, k_{t+1}, d_{t+1}^U) + \sigma V_{t+1}^U(h_{t+1}, k_{t+1}, d_{t+1}^U, b_{t+1}^0)\right)\Big) \\ & + g_t \tilde{V}_{t+1}^S(h_{t+1}, k_{t+1}, d_{t+1}^U, d_{t+1}^S, r_{t+1}, 1, b_{t+1}^0) \Big| h_t\Bigg] \Bigg\} \end{aligned} \tag{18}$$

$$\text{s.t.}$$

$$c_t \quad = \quad y_t = (1-\tau) w_t(k_t). \tag{19}$$

Value of being on sickness leave

$$
\begin{aligned}
V_t^S(h_t, k_t, d_t^U, d_t^S, r_t, o_t, b_t^0) \quad = \quad & \max_{e_t \in \{0,1\}} \Bigg\{ \eta \frac{c_t^{1+\mu}}{1+\mu} + \beta E \Bigg[(1-e_t)\big(\mathbf{1}_{\{o_t=1\}}(\sigma V_{t+1}^S(h_{t+1}, k_{t+1}, d_{t+1}^U, d_{t+1}^S, 0, b_t^0) \\
& +(1-\sigma) V_{t+1}^S(h_{t+1}, k_{t+1}, d_{t+1}^U, d_{t+1}^S, o_{t+1}, b_t^0)) + e_t\big(\mathbf{1}_{\{o_t=1\}}\big((1-\sigma) V_{t+1}^E(h_{t+1}, k_{t+1}, \\
& +\sigma V_{t+1}^U(h_{t+1}, k_{t+1}, d_{t+1}^U, b_{t+1}^0)\big) + \mathbf{1}_{\{o_t=0\}} V_{t+1}^U(h_{t+1}, k_{t+1}, d_{t+1}^U, b_{t+1}^0)\big) \big| h_t \Bigg] \Bigg\} \\
\text{s.t.} & \\
c_t \quad = \quad & y_t = (1-\tau) b_t^S(r_t).
\end{aligned}
$$

3.4 Solve

Backwards induction
Discretize health grid with the Tauschen method

3.5 Simulate

The two types have different initial health level to match that workers with lower attachment to the labor market have worse health empirically.

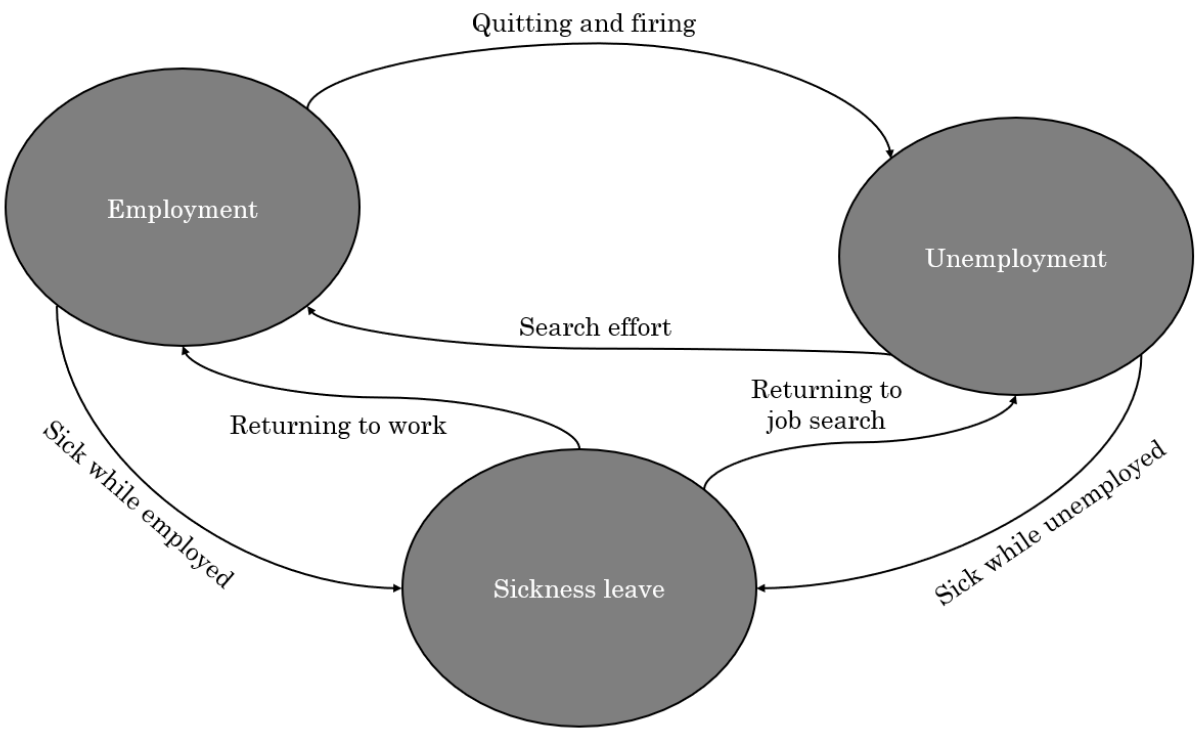

Figure 2: Model flow to and from employment, unemployment and sickness leave

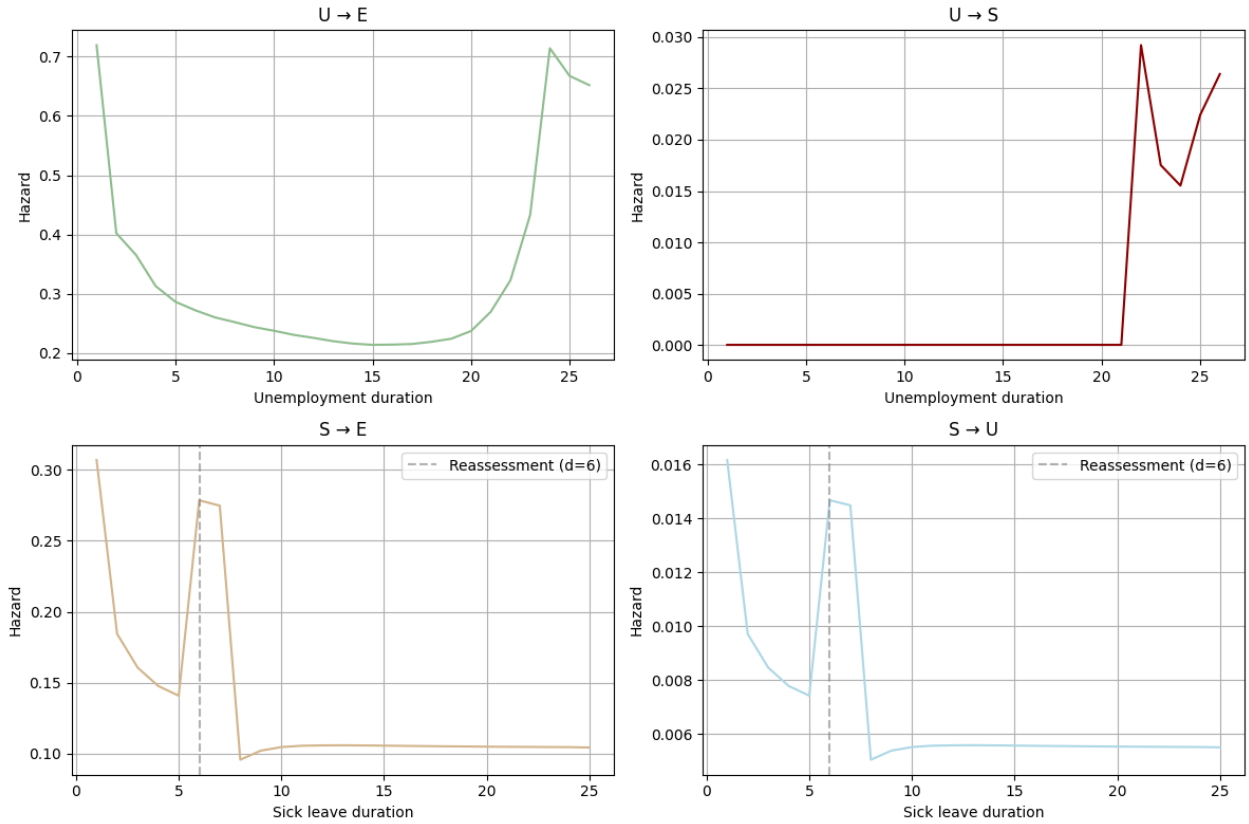

Figure 3: First model results

4 Structural estimation

4.1 Identification

4.2 Results

I just tried estimating it on an old hazard rate I had.

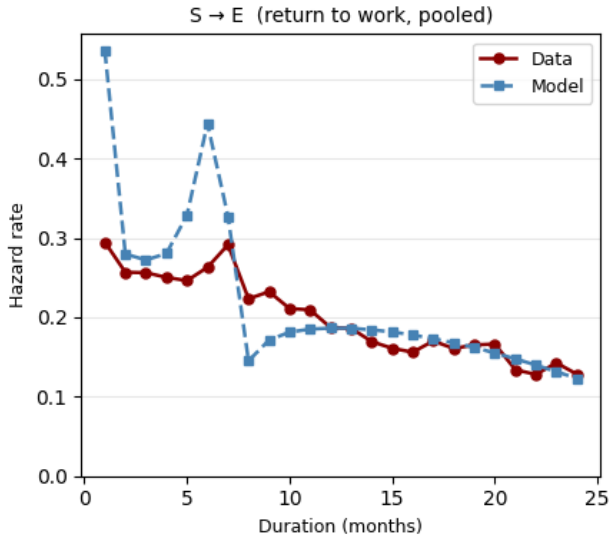

Figure 4: First results

5 Discussion and concluding remarks

References

Capatina, E. (2015). Life-cycle effects of health risk. *Journal of Monetary Economics*, *74*, 67–88. https://doi.org/10.1016/j.jmoneco.2015.06.002

Capatina, E., & Keane, M. (2023). Health Shocks, Health Insurance, Human Capital, and the Dynamics of Earnings and Health.

De Nardi, M., Pashchenko, S., & Porapakkarm, P. (2025). The Lifetime Costs of Bad Health. *Review of Economic Studies*, *92*(3), 1987–2026. https://doi.org/10.1093/restud/rdae080

DellaVigna, S., Lindner, A., Reizer, B., & Schmieder, J. F. (2017). Reference-Dependent Job Search: Evidence from Hungary*. *The Quarterly Journal of Economics*, *132*(4), 1969–2018. https://doi.org/10.1093/qje/qjx015

French, E. (2005). The Effects of Health, Wealth, and Wages on Labour Supply and Retirement Behaviour. *The Review of Economic Studies*, *72*(2), 395–427. https://doi.org/10.1111/j.1467-937X.2005.00337.x

Hosseini, R., Kopecky, K. A., & Zhao, K. (2022). The evolution of health over the life cycle. *Review of Economic Dynamics*, *45*, 237–263. https://doi.org/10.1016/j.red.2021.07.001

Jeske, K., & Kitao, S. (2009). U.S. tax policy and health insurance demand: Can a regressive policy improve welfare? *Journal of Monetary Economics*, *56*(2), 210–221. https://doi.org/10.1016/j.jmoneco.2008.12.004

Keane, M. P. (2016). Life-cycle Labour Supply with Human Capital: Econometric and Behavioural Implications. *The Economic Journal*, *126*(592), 546–577. https://doi.org/10.1111/ecoj.12363

Khwaja, A. (2010). Estimating willingness to pay for medicare using a dynamic life-cycle model of demand for health insurance. *Journal of Econometrics*, *156*(1), 130–147. https://doi.org/10.1016/j.jeconom.2009.09.011

Kitao, S. (2014). A life-cycle model of unemployment and disability insurance. *Journal of Monetary Economics*, *68*, 1–18. https://doi.org/10.1016/j.jmoneco.2014.07.008

6 Appendix